THE NAIVE COTANGENT COMPLEX OF A COLIMIT OF RING MAPS IS THE COLIMIT OF THE NAIVE COTANGENT COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The naive cotangent complex as a two-term complex from a presentation. Recalled at the start of the counted proof (lines 35782-35784). Describes the complex in terms of the kernel of the canonical presentation, which is what can be compared along the colimit.
- Colimits commute with polynomial algebras and with kernels. Used at the end of the counted proof (lines 35785-35787). Identifies the polynomial algebra and the kernel of the presentation over the colimit with the colimits of the corresponding objects over the index set.

Lemma 0.1. *Let $R_\lambda \to S_\lambda$ be a system of ring maps over the directed set Λ. Set $R = \operatorname{colim} R_\lambda$ and $S = \operatorname{colim} S_\lambda$. Then $NL_{S/R} = \operatorname{colim} NL_{S_\lambda/R_\lambda}$.*

Proof. Recall that $NL_{S/R}$ is the complex $I/I^2 \to \bigoplus_{s \in S} S\mathrm{d}[s]$ where $I \subset R[S]$ is the kernel of the canonical presentation $R[S] \to S$. Now it is clear that $R[S] = \operatorname{colim} R_\lambda[S_\lambda]$ and similarly that $I = \operatorname{colim} I_\lambda$ where $I_\lambda = \operatorname{Ker}(R_\lambda[S_\lambda] \to S_\lambda)$. Hence the lemma is clear. □

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